

On the Long-Run Holding Returns of Japanese Stocks: Individual Stocks vs. Portfolios^{*) **)}

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Abstract

We investigate the long-run holding returns of the stocks listed on eight Japanese stock exchanges with weekly return data from 1977 through 2007. We find existence of significant positive autocorrelations for the smallest and the middle quintile portfolios from the variance ratio test, but not for the largest. However, returns are negatively correlated at the individual stock level and positively correlated at the portfolio level. We demonstrate that the negative correlation disappears as the number of stocks in each portfolio is increased to four, which shows the idiosyncratic risk of individual stock is dominated by the positive autocorrelation at the portfolio level.

I. Introduction

In the current paper we test whether long-term stock returns in the Japanese stock markets follow a random walk or not. We focus on firms listed on the Japanese eight stock exchanges, and we employ observations for stock returns from 1977 through 2007. We test the random walk hypothesis of stock returns using both the broadest and longest observations.

We investigate whether both autocorrelations and variance ratios of weekly stock returns in Japanese markets from long-term observations are different at 1) the individual stock level, 2) the market portfolio level and 3) the size ranked portfolio level by employing two types of variance ratio tests. The first test is based on an estimator developed by Lo and MacKinlay (1988) and Campbell et al. (1997), and the second test is based on an estimator by Kan (2006). Moreover, we vary the number of stocks included in the portfolios and analyze the reasons for the conflicting results found at the portfolio level and the individual stock level. This research is the most extensive for testing of random walk hypothesis for stock returns in Japanese markets.

Section two surveys the literature. Section three motivates our study. Section four explains our data. Section five presents our main result from the variance ratio test and investigates the relation between variance ratios and the number of stocks. Section six concludes.

II. Literature

Poterba and Summers (1988) and Lo and MacKinlay (1988) empirically found that

variance ratios from stock returns for U.S. companies are statistically different from one, refuting the random walk hypothesis by Fama (1970). That is, Lo and MacKinlay (1988) find that variance ratios for equal weighted market portfolio returns are statistically greater than one for 2, 4, 8, and 16 weeks. They also find that variance ratios are higher for stocks with smaller market capitalization than those with larger market capitalization. Lo and MacKinlay (1999) and Campbell, Lo, and MacKinlay (1997) updated their observations and showed the robustness of their evidence using weekly return observations. On the other hand, Campbell, Lo, and MacKinlay (1997) find that the variance ratio is marginally larger than one for the value weighted return and it is slightly smaller than one at the individual stock level.

Many researchers used test statistics from Lo and MacKinlay (1988) and conducted a variance ratio test for stock returns from various international markets.¹ However, there are few papers that investigate the random walk nature of Japanese stock returns and the results are mixed. For example, using weekly TOPIX observations, Iwaisako (2007) does not reject the random walk hypothesis. Using the weekly MSCI Japan index, Patro and Wu (2004) do not reject the random walk hypothesis either. Iwaisako (2007) finds that three daily size-ranked indices from the Tokyo Stock Exchange follow a similar cross-autocorrelation pattern as in Lo and MacKinlay (1988). However, he also finds that TOPIX (whose returns are value weighted) does not exhibit any statistically significant

¹ See evidence by Frennberg and Hansson (1993) for Sweden, Chun (2000) for Hungary, Buguk and Brorsen (2003) for Turkey, Laopodis (2004) for Greece, Depenya and Gil-Alana (2007) for Spain, Lagoarde-Segot and Lucey (2008) for middle-eastern north Africa, Ayadi and Pyun (1994) for Korea, Hoque, Kim and Pyun (2007) for 8 Asian countries, Kim and Shamsuddin (2007) for 10 Asian countries, and Patro and Wu (2004) for international stock indices.

positive autocorrelation patterns, and he attributes this finding to the dominance of large-cap stocks in TOPIX compared with those in the CRSP index. Recently, Chou et al. (2007) have also found the evidence that contrarian strategies are profitable for Japanese stocks. They claim that the profit comes from cross-autocorrelations of abnormal returns from Fama and French three factor model.

Given conflicting evidence for the random walk hypothesis as well as the return reversal vs. continuation controversy on Japanese stock returns, our study sheds light on this important topic in identifying stock return generating processes in Japan.

III. Test of the Random Walk Hypothesis

In this paper we compute the variance ratio and test the random walk hypothesis using overlapping observations.² For this purpose, we consider the following equation (1) wherein $\bar{\sigma}_q^2$ denotes the sample variance per week (the weekly sample variance divided by the number of the weeks used, q), T the number of the observations, and $\mathbf{R}$ ($T \times 1$) the row vector of returns.

² It is well known that, in an overlapped data like ours, the estimated variance ratio is downward biased in the small sample (Lo and MacKinlay, 1989). Tse, Ng, and Hang (2004) propose the statistics to correct for this bias. Also, Poterba and Summers (1988) and Richardson and Stock (1989) discuss the possible loss of the power of test when overlapping data is used. Because our data covers the long years with weekly observations with minimum number of lag coefficients, we believe that this overlapping problem is kept to the minimum. However, we plan to correct for this problem in our future study.

$$(1) \quad \hat{\theta}(q) \equiv \frac{\bar{\sigma}_q^2}{\bar{\sigma}_1^2} = \frac{T-1}{m} \left(\frac{\mathbf{R}' \mathbf{M}' \mathbf{H}' \mathbf{H} \mathbf{M} \mathbf{R}}{\mathbf{R}' \mathbf{M} \mathbf{R}} \right), \quad q=2,3,\dots,$$

where

$$n \equiv T-q+1, m \equiv q n(n-1) \frac{1}{T}, \mathbf{M} \equiv \mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T', \mathbf{H}_q \equiv \begin{bmatrix} 1 \dots 1 & 0 & \dots & 0 \\ 0 & 1 \dots 1 & 0 & \dots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & 0 & 1 \dots 1 \end{bmatrix}.$$

In equation (1), $\mathbf{I}_T$ is a unit matrix with size $T \times T$ where all elements are one, $\mathbf{1}_T$ is a $T \times 1$ column vector where all elements are one, and $\mathbf{H}_q$ is a matrix with $T \times n$ in which i -th row and $(i+q-1)$ -th columns are filled with ones where the number of the elements is q , and zeros elsewhere. Because returns are measured in natural logarithms, if there is a positive (negative) autocorrelation in the observations, the estimated variance ratio should become larger (smaller) than one. The null hypothesis for the existence of the random walk is that the variance ratio should equal one.

We adopt two types of tests for $H_0: \theta(q) = 1$ against $H_1: \theta(q) \neq 1$ for all q in this paper. The first test is based on an asymptotic distribution under a heteroskedastic assumption by Lo and MacKinlay (1988):

$$(2) \quad \Pr[\hat{\theta}(q) < c] = \int_{-\infty}^c \frac{1}{\sqrt{2\pi s}} \exp\left[-\frac{1}{2} \left(\frac{c-1}{s}\right)^2\right] dc,$$

where $s^2 = \sum_{i=1}^k \left(\frac{2(q-i)}{q}\right)^2 \left(1 + \frac{\overline{\text{Cov}}(\hat{\varepsilon}_t^2, \hat{\varepsilon}_{t-i}^2)}{\hat{E}[\hat{\varepsilon}_t^2]^2}\right), \hat{\varepsilon}_t = r_t - \hat{\mu},$

The second test is based on the exact distribution under a homoskedastic assumption by Kan (2006):

$$(3) \quad \Pr[\hat{\theta}(q) < c] = \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \left[\frac{\sin\left(\frac{1}{2} \sum_{i=1}^{T-1} \arctan(2x\lambda_i)\right)}{x \prod_{i=1}^{T-1} (1+4x^2\lambda_i^2)^{1/4}} \right] dx, \text{ where } \lambda_i = d_i - \frac{cm}{T-1}.$$

In equations (2) and (3), $\hat{\mu}$ is the sample mean, and the sequence $\{d_i\}$ is composed of $(q-2)$ zeros and n eigenvalues from the matrix, $\mathbf{H}_q \mathbf{H}_q' - (q^2/T) \mathbf{1}_n \mathbf{1}_n'$. Whenever $\overline{\text{Cov}}(\hat{\varepsilon}_t^2, \hat{\varepsilon}_{t-i}^2) = 0$ holds for $\forall i$, the equation (2) above degenerates to the case of homoskedasticity, where $s^2 = 2(k-1)(2k-1)/3k$ as $T \rightarrow \infty$ while holding k fixed.³ We consider the case where q is an integer larger than one and less than 260. In other words, we focus not only on the short term autocorrelations within a year but on the long term ones.

IV. Data

Our study is the first study to employ weekly individual stock return observations for Japan and investigates autocorrelations and variance ratios for individual stock returns in Japanese markets. We emphasize that no previous studies exist in the field from this perspective and with this data set. We conjecture the comparison between U.S. and

³ Kan (2006) and Kan and Wang (2010) report simulation results and concludes that with large lags (q) his model is more accurate than Lo and MacKinlay (1988) even if there is heteroskedasticity in the data.

Japanese stock returns will also add further insights into the characteristics of stock returns in well developed capital markets.

Our observation period starts from August 31st 1977 and ends in June 27th 2007. The sample consists of all the firms listed on eight Japanese stock exchanges (Tokyo, Osaka, Nagoya, Fukuoka, Sapporo, Niigata, Kyoto, and Hiroshima). The data source is the Nikkei Portfolio Master Daily Return file provided by Nikkei Media Marketing.

From the daily observation on stock prices, we compute weekly returns between a Wednesday close price and the next Wednesday close price. When a particular Wednesday turns out to be a non-trading day, we use the Thursday close price and if it is not available, we use the Tuesday close price. When neither price is available, we skip the week and do not include it in our observations.⁴ With this data construction method, we obtain 1,541 weekly time-series return observations from August 31st 1977 to June 27th 2007. Although the original database contained both foreign and Japanese firms listed on the Japanese stock exchanges, our final sample contains only Japanese firms. We also split the observations on August 26th 1992 so that the first and the second sub sample periods contain an observation of 772 and 769 weeks in order to test for the robustness of our result.

V. Variance Ratios and Autocorrelation Patterns

A. Individual Stock Returns

In Table 1, we show the characteristics of variance ratios at the individual stock level. The total number of stocks is 1,147 for which the complete return history, without any

⁴ This sampling scheme is to minimize the weekend and no-trading day effects of the public and private information flow into the stock market (French and Roll, 1986).

missing observations, is available between August 31st 1977 and June 27th 2007 (1541 consecutive weeks). We also form quintile portfolios by market capitalization at the middle of the sample period (August 26th 1992)⁵.

Table 1 about here

After computing 1,147 variance ratios for all the 1,147 stocks from each of its stock returns, we calculate the arithmetic mean and median of these variance ratios. Remarkably, we find that both the mean and the median of all the variance ratios are less than one. As we report in the next sub-section, the variance ratios at the portfolio level are greater than one in most of the cases, so the behavior of individual stock returns are remarkably different from those of portfolio returns.⁶ Judging from variance ratios, autocorrelations are negative for most cases of the individual stocks.⁷

For the case of variance ratios computed from all 1,147 firms denoted as “All stocks” in Table 1, mean variance ratios are 0.95, 0.93, 0.91, 0.87, 0.82, 0.84 and 0.83 for lags of 2, 4, 8, 26, 52, 156 and 260 weeks, respectively. They are all below one. In the next three rows in Table 1, we report variance ratios calculated from five quintiles each of which contains

⁵ We obtain almost the same results even if we sort the stocks at the beginning or at the end of the sample period. The results are available upon request from the authors.

⁶ We emphasize the results for index portfolios formed from the sample of 1,147 firms where this subset of firms does not indicate any different characteristics from our total sample (which was reported in Table 2). We also used the same sample for individual stock estimations and they do not suffer from the survivorship bias.

⁷ Lo and MacKinlay (1988) also report a similar finding for U.S. data at the individual stock level. In private conversation with Sheridan Titman, he mentioned a similar finding of the behavior of value stocks *versus* growth stocks for the U.S., in which a different return performance is observed among portfolios *versus* individual stocks.

229 stocks. We form the quintile by sorting each stock by the market capitalization on August 26, 1992.⁸ We report only the variance ratios for the smallest, the middle and the largest size-ranked quintile portfolios. We find that variance ratios still stay below one regardless of the size, so there is a negative autocorrelation at the individual stock level. We also note that stocks in the largest quintile have, on average, smaller autocorrelations than those in the smallest quintile for longer horizons; i.e., for the three year horizon, 0.96 vs. 0.77 and for the five year horizon, 0.98 vs. 0.70 for the mean. The result shows that the characteristics of returns from firms with larger market capitalization are closer to a random walk for the time length of more than three years.

Figure 1 about here

Figure 1 plots the median values of variance ratios for weekly individual stocks in Table 1 by adding the observations for 104 and 208 weeks not reported in Table 1. The median for stocks in the smallest quintile stays unanimously below one and tends to decline. On the other hand, the median for stocks in the largest quintile increases between the second and the fourth year. Finally, the tendency of negative correlations is much stronger for stocks in the smallest quintile than for stocks in the middle and largest quintiles.

B. Market Index Returns

⁸ We also formed quintile portfolio by sorting on both the first day of the observations and the last day of the observations and obtained similar results. The results are available from the author upon request.

Table 2 shows the summary of sample statistics for both equal-weighted and value-weighted market portfolios for the following three periods: 1) the whole period from August 31st 1977 to June 27th 2007 (1,541 weeks), 2) the first sub-period from August 31st 1977 to August 26th 1992 (772 weeks) and 3) the second sub-period from August 26th 1992 to June 27th 2007 (769 weeks).

Panel A shows the result for the equal-weighted portfolio and Panel B shows the results for the value-weighted portfolio. Value-weighted portfolios are weighted by the market capitalization as of August 26th, 1992.⁹

Table 2 about here

The pattern of autocorrelations for different lags is remarkably different between the equal-weighted portfolio in Panel A and the value-weighted portfolio in Panel B. All the autocorrelation coefficients of the equal-weighted portfolio in Panel A are significantly positive up to the first lag. Ljung-Box Q statistics to the fourth lags and to the eighth lags are also significant. On the other hand, in the case of the value-weighted portfolio in Panel B, only the coefficients for the second lag are significantly positive for the total period and the first sub-period. None of other coefficients are significant. Ljung-Box Q statistics are significant only to the fourth lags and to the eighth lags for the first sub-period. They are

⁹ We did not correct for treasury stocks which should strictly be subtracted from outstanding shares. The treasury stock program was only legalized in 2001 by the Security Exchange Code in Japan. The most active firms which buy back their own stock is Toyota Motor Co. and NTT Docomo Co. and they have an inventory of about 11% and 5% Treasury stock, respectively. Note the overall averages across firms are much smaller and most firms' figures are less than 1% in 2006.

not significant for the total period and the second sub-period.

In Table 3, we report variance ratios for the same value and equal weighted portfolios for the same three periods as in Table 2. We show results based on two different tests in parentheses and brackets. The cumulative probability inside the parenthesis corresponds to the asymptotically consistent statistics by Lo and MacKinlay (1988) and the one inside the brackets corresponds to the statistics derived by Kan (2006).¹⁰ Panel A shows that variance ratios for equal-weighted portfolios from the total period and the first sub-periods are all significantly greater than one either at the 1 per cent level or at the 5 per cent level.¹¹ Variance ratios from the second sub-periods are also significantly greater than one except for the 52, 156 and 260 week cases. The magnitude of variance ratios keeps increasing for the entire period and more rapidly for the first sub-period, while it is relatively stable for the second sub-period.

Table 3 about here

However, the variance ratios for the value-weighted portfolio demonstrate remarkably different patterns. In panel B, we find that variance ratios from the total period are not

¹⁰ In order to further confirm the statistical significance of autocorrelation structures of index returns, we also conducted a version of the ARCH (Engel, 1982) test proposed by Lo and MacKinlay (1988). Lo and MacKinlay (1989) also report the simulation results to show that their test statistics are equipped with stronger power than the Dickey-Fuller test and the Box-Pierce Q test. The estimated Chi-square values for our data were 75.1 and 48.2 for the equal weighted index and the value weighted index, respectively. We find that all these values are significant at the 5% level, thus the result suggests the existence of a positive autocorrelation structure from the ARCH test as well.

¹¹ This observation is mostly consistent with the previous finding for the equal weighted indices for the U.S. as reported in Lo and MacKinlay (1988).

significantly greater than one for all the lags except for 260 weeks. Variance ratios from the first sub-period are not greater than one except for the lag of 156 and 260 weeks.¹² The variance ratios from the second sub-period are not significantly different from one. Again, the magnitude of variance ratios keeps increasing for the entire period and more rapidly for the first sub-period, but it is relatively stable for the second sub-period.

Overall, we conclude that a random walk hypothesis is rejected for the equal-weighted portfolio in most of the cases, while it is rejected for the value-weighted portfolio only for longer lags to a weaker extent. Note we previously indicated that variance ratios for most of the individual stocks in Table 1 are less than one, while we find that all variance ratios for equal weighted portfolios in the total period and the first sub-period in Table 3 are greater than one.

Accordingly, this discrepancy between variance ratios at the individual level and ones at t portfolio levels need to be investigated in the context of portfolio formation and its return profile. We focus on the equal-weighted portfolios because the variance ratios are significantly greater than one for most of the holding periods and also because variance ratios for the value-weighted portfolios may be affected by stocks with large market capitalizations. In the next sub-section, we investigate when the change of sign occurs as we change the number of stocks in the portfolio.

C. Size-sorted Portfolio Returns

Next, we form five portfolios based on the market capitalization of the firm and

¹² These results are also consistent with the findings reported in Campbell, Lo, and MacKinlay (1997) for U.S. data with respect to the value weighted index.

investigate if the pattern of autocorrelations of equal-weighted portfolio from larger firms is different from smaller firms. Specifically, we form five equal-weighted portfolios based on size as measured by the total market value on August 26th 1992. We consider the same three sample periods as in Table 2 and 3. In Table 4, we report the summary statistics for the largest, the middle, and the smallest quintile portfolios, each of which contains equally-weighted stocks.¹³

We find that the autocorrelation coefficients from the total and sub-sample periods are positive and significant for the smallest portfolio in Panel A between the first and third lag. In Panel A (i), the autocorrelation coefficients for the smallest stocks are 0.236, 0.118 and 0.087 for lag 1, 2 and 3, respectively, for the total period. Ljung-Box Q statistics from total and sub-sample periods for this smallest portfolio are also significant at the 1 % level up to the lags of 4 and 8. The magnitude of variance ratios keeps increasing for the entire period and more rapidly for the first sub-period, but it is relatively stable for the second sub-period. Autocorrelation coefficients from total and sub-sample periods are also positive and significant for the middle portfolio in Panel B for the lag of one week. Again, the magnitude of variance ratios keeps increasing for the entire period and more rapidly for the first sub-period, but it is relatively stable for the second sub-period.

Table 4 about here

However, most of the autocorrelation coefficients are not significant for the largest

¹³ Complete results are available upon request from the authors.

quintile portfolio in Panel C. The magnitude of variance ratios keeps increasing for the entire period and more rapidly for the first sub-period, but it is relatively stable for the second sub-period. Thus, the behavior of each quintile portfolio is quite different from the standpoint of the significance of the variance ratio but similar from the standpoint of the pattern of the magnitude, which is a new finding.

Table 5 about here

Table 5 reports the variance ratios for the same three set of size-sorted and equal-weighted portfolios in Table 4. We find that variance ratios are statistically greater than one for the smallest quintile portfolios in most of the cases and for the middle quintile in many of the cases. However, this is not the case for the largest quintile portfolio. In the case of the smallest portfolios for the total period in Panel A, the variance ratios are 1.24, 1.52, 1.74, 2.09, 2.11, 2.43, and 2.27 for the lags of 2, 4, 8, 26, 52, 156 and 260 weeks, respectively. Note that variance ratios are all significant at the 1% level. On the other hand, in the case of largest portfolios for the total period, they are not significant except for the lag of 260 weeks as shown in Panel C even though the estimated ratios exceed one.¹⁴ This last finding augments the previously found result on the autocorrelation structure for longer week returns among the large stocks in Table 4.

¹⁴ Our findings are similar to the one for the U.S. by Campbell, Lo, and MacKinlay (1997). Moreover, Lo and MacKinlay (1988) show weaker evidence that the random walk hypothesis is rejected for large stocks.

D. Relation between the Number of Stocks and Variance Ratios

In the previous sub-sections we demonstrated evidence that 1) the equal weighted market portfolio returns, 2) the value weighted market portfolio returns to a lesser extent, 3) the size-sorted portfolio return from the smallest quintiles and 4) the size-sorted portfolio return from the middle quintiles to a lesser extent are positively correlated while individual stock returns are negatively correlated. In this sub-section, we investigate how the above results depend on the number of stocks in the portfolio and how the individual idiosyncratic risk begins to be dominated by the positive autocorrelations at the portfolio level. In order to find when a variance ratio exceeds one, we gradually increase the number of stocks contained in size-sorted and equal-weighted portfolios.

There are several ways to sort the individual stocks. In this section, we sort based on the size, as portfolio sorted by size is often used in empirical analyses.¹⁵

Table 6 reports the relation between the number of stocks in the portfolio and variance ratio for each of these portfolio. Table 6 shows the variance ratios for portfolios which consist of one to 1,147 stocks. Although we find in sub-section 4.1 that the median of variance ratios for individual stocks stays below one, the median of variance ratios for size-sorted portfolios that consist of only three stocks already exceeds one.

Table 6 about here

For portfolios with 22 or more equal-weighted stocks, the variance ratios keep increasing

¹⁵ In other times portfolios are sorted by industries, book to market or pre-betas.

with a modest one-time decline at 52 weeks. For portfolios with stocks between three and five, they keep rising more slowly again with a modest one-time decline at 52 weeks. For portfolios with two stocks, they stay around the value of one and for portfolios with a single stock, they stay below one and even decline steadily over five years.

Figure 2 shows the median values of variance ratios from two different portfolios. First, we form 229 size-sorted portfolios each with five stocks. We divide these 229 portfolios into five quintiles. The smallest contains 46 portfolios, the second smallest contains 46 portfolios, the middle contains 45 portfolios, the second largest contains 46 portfolios and the largest contains 46 portfolios. Finally we obtain the median value of variance ratios from the 229 portfolios and ones from three of quintile portfolios.

The median values calculated from 1) 1,147 individual stocks and 2) each of 229 individual stocks lie below one in Figure 1. However, the same statistics calculated from 1) 229 portfolios which contain only five stocks and 2) three quintile portfolios which contain five stocks in Figure 2 take values greater than one. In comparison with the strongly negative autocorrelations for the smallest quintile stocks, we also find that the tendency of positive autocorrelations is much stronger for smallest quintile portfolios than for middle and largest quintile portfolios except for 260 weeks.

Figure 2 about here

Finally, we can reconcile the negative autocorrelation phenomenon at the individual level and the positive autocorrelation observation at the portfolio level. From the viewpoint of

practical portfolio management, wherein the stocks of the typical portfolio would be normally larger than three, the positive autocorrelation dominates and we obtain evidence of the price continuation for Japanese stocks. Moreover, by comparing Figures 1 and 2, we find that the negative autocorrelation of the small sized stocks at the individual level demonstrate a quick turnaround to the positive autocorrelation as soon as small sized stocks are formed into portfolios. These are new findings and it is another contribution of the paper.¹⁶

VI. Conclusion

We investigated the autocorrelation structure of all the listed firms in Japan using weekly return data from 1977 through 2007. We rejected the random walk hypothesis for equal weighted portfolios, smallest and middle size ranked portfolios, by utilizing both the asymptotically consistent and exact variance ratio statistics, but not for the large size ranked portfolio. We also confirm that the price variations of larger firms lead those of smaller firms.

However, while there are positive autocorrelations for these portfolios, the correlations are negative at the individual level. Then, we demonstrated that negative signs turn into positive ones as soon as the number of stocks increases to four, which implies that the individual idiosyncratic risk of the stocks are dominated by the positive autocorrelation at the portfolio level.

¹⁶ However, once we consider the possibility of rebalancing the portfolio on a particular date, bid and ask bounce problems may emerge and substantially affect our results. Overcoming this potential problem is our ongoing work.

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TABLE 1
Summary Statistics of Variance Ratios for Weekly Individual Stocks

		<i>q</i> -weeks						
		2	4	8	26	52	156	260
All stocks <i>N</i> =1,147	average	0.95	0.93	0.91	0.87	0.82	0.84	0.83
	median	0.95	0.93	0.90	0.84	0.78	0.78	0.75
	%(VR<1)	86.14%	83.96%	82.82%	81.60%	82.74%	76.81%	73.58%
Smallest quintile <i>N</i> =229	average	0.94	0.91	0.88	0.85	0.81	0.77	0.70
	median	0.94	0.91	0.88	0.82	0.76	0.74	0.64
	%(VR<1)	88.21%	89.96%	84.72%	84.28%	85.59%	82.53%	85.15%
Middle quintile <i>N</i> =229	average	0.96	0.94	0.92	0.87	0.81	0.82	0.82
	median	0.96	0.94	0.90	0.85	0.78	0.76	0.75
	%(VR<1)	84.72%	79.04%	79.91%	80.35%	87.34%	81.22%	75.98%
Largest quintile <i>N</i> =229	average	0.95	0.93	0.92	0.89	0.89	0.96	0.98
	median	0.95	0.93	0.92	0.87	0.86	0.88	0.91
	%(VR<1)	85.59%	85.15%	82.10%	80.79%	71.62%	61.57%	58.08%

All values in the table are calculated from variance ratios for weekly individual stock returns with complete return histories from August 31, 1977 to June 27, 2007 ($T=1,541$). %(VR<1)'s represent percentages where variance ratios are less than one.

TABLE 2
Summary Statistics for Weekly Market Portfolio Returns

		Sample		Std.	Autocorrelations (ρ_q)					
Time period		Size	Mean	Dev.	$q = 1$	2	3	4	Q_4	Q_8
Panel A. Equal-weighted portfolio (All stocks)										
(i)	77/8/31 - 07/6/27 $T=1,541$	1,541	0.3%	2.5%	14.3%**	9.1%**	4.1%	-0.8%	46.9**	49.6**
(ii)	77/8/31 - 92/8/26 $T=772$	772	0.3%	2.2%	11.8%**	15.7%**	5.5%	0.4%	32.4**	47.9**
(iii)	92/8/26 - 07/6/27 $T=769$	769	0.2%	2.9%	13.9%**	5.1%	5.3%	-1.3%	19.3**	24.5**
Panel B. Value-weighted portfolio (All stocks)										
(i)	77/8/31 - 07/6/27 $T=1,541$	1,541	0.2%	2.3%	-0.4%	5.9%*	1.7%	-1.8%	6.3	10.2
(ii)	77/8/31 - 92/8/26 $T=772$	772	0.3%	2.2%	-3.6%	10.4%*	2.6%	-2.2%	10.3*	17.1*
(iii)	92/8/26 - 07/6/27 $T=769$	769	0.2%	2.5%	0.6%	1.4%	2.7%	-1.6%	1.0	3.7

All values in the table are calculated from variance ratios for weekly individual stock returns with complete return histories from August 31, 1977 to June 27, 2007 ($T=1,541$). Q is Ljung-Box statistics. * and ** indicate significance at the 5% and 1% level, respectively.

TABLE 3
Variance Ratio test for Weekly Market Portfolio Returns

	$q = 2$	4	8	26	52	156	260
Panel A. Equal-weighted portfolio							
(i) 77/8/31 - 07/6/27	1.14	1.33	1.44	1.59	1.52	1.95	2.13
$T=1,541$	(0.001) ** [0.000] **	(0.000) ** [0.000] **	(0.000) ** [0.000] **	(0.003) ** [0.001] **	(0.032) * [0.018] *	(0.015) * [0.029] *	(0.018) * [0.046] *
(ii) 77/8/31 - 92/8/26	1.11	1.35	1.48	1.72	2.07	3.17	3.08
$T=772$	(0.031) * [0.000] **	(0.003) ** [0.000] **	(0.006) ** [0.000] **	(0.012) * [0.003] **	(0.006) ** [0.005] **	(0.000) ** [0.011] *	(0.005) ** [0.028] *
(iii) 92/8/26 - 07/6/27	1.14	1.28	1.41	1.50	1.19	1.26	1.28
$T=769$	(0.030) * [0.000] **	(0.009) ** [0.000] **	(0.012) * [0.000] **	(0.066) [0.019] *	(0.594) [0.243]	(0.641) [0.258]	(0.676) [0.254]
Panel B. Value-weighted portfolio							
(i) 77/8/31 - 07/6/27	1.00	1.06	1.10	1.18	1.20	1.63	1.87
$T=1,541$	(0.545) [0.565]	(0.188) [0.103]	(0.173) [0.098]	(0.166) [0.119]	(0.212) [0.170]	(0.061) [0.073]	(0.045) * [0.074]
(ii) 77/8/31 - 92/8/26	0.97	1.06	1.09	1.22	1.53	2.49	2.58
$T=772$	(0.720) [0.833]	(0.294) [0.179]	(0.293) [0.187]	(0.225) [0.148]	(0.086) [0.059]	(0.006) ** [0.027] *	(0.015) * [0.049] *
(iii) 92/8/26 - 07/6/27	1.01	1.04	1.09	1.12	0.89	0.81	0.81
$T=769$	(0.446) [0.432]	(0.327) [0.290]	(0.236) [0.195]	(0.297) [0.263]	(0.640) [0.603]	(0.644) [0.552]	(0.615) [0.486]

All values in the table are calculated from variance ratios for weekly portfolio returns with complete individual return histories from August 31, 1977 to June 27, 2007 ($T=1,541$). The values within parentheses are p-values from Lo and MacKinlay's (1988) heteroskedastic consistent test. The values within square brackets are p-values from Kan's (2006) exact test under the elliptical distribution assumption. * and ** indicate that the corresponding variance ratios are statistically different from 1 at the 5% and 1% significance level, respectively.

TABLE 4
Summary Statistics for Weekly Size-sorted Portfolio Returns with Equal Weights

		Sample		Std.	Autocorrelations (ρ_q)					
Time period		Size	Mean	Dev.	$q = 1$	2	3	4	Q_4	Q_8
Panel A. Portfolio of firms with market values in the smallest quintile (rank 1)										
(i)	77/8/31 - 07/6/27	1,541	0.3%	2.8%	23.6%**	11.8%**	8.7%**	1.3%	119.6**	123.3**
(ii)	77/8/31 - 92/8/26	772	0.3%	2.4%	27.6%**	16%**	12.2%**	5.3%	92.4**	102.4**
(iii)	92/8/26 - 07/6/27	769	0.3%	3.2%	20.2%**	10.4%**	8.5%**	-0.1%	45.4**	48.8**
Panel B. Portfolio of firms with market values in the middle quintile (rank 3)										
(i)	77/8/31 - 07/6/27	1,541	0.2%	2.8%	10.9%**	8.2%**	2.9%	-1.6%	30.3**	32.5**
(ii)	77/8/31 - 92/8/26	772	0.3%	2.4%	7.8%*	14.6%**	3.7%	-0.9%	22.3**	38**
(iii)	92/8/26 - 07/6/27	769	0.2%	3.2%	11.1%**	4.1%	4.3%	-2.0%	12.5*	18.2*
Panel C. Portfolio of firms with market values in the largest quintile (rank 5)										
(i)	77/8/31 - 07/6/27	1,541	0.2%	2.3%	-0.1%	5.6%*	1.8%	-1.6%	5.8	8.5
(ii)	77/8/31 - 92/8/26	772	0.3%	2.2%	-2.8%	9.9%*	3.3%	-1.6%	9.2	15.5
(iii)	92/8/26 - 07/6/27	769	0.2%	2.5%	0.4%	1.4%	2.6%	-1.8%	0.9	3.2

All values in the table are calculated from variance ratios for weekly individual stock returns with complete return histories from August 31, 1977 to June 27, 2007 (T=1,541). Q is Ljung-Box statistics. * and ** indicate significance at the 5% and 1% level, respectively.

TABLE 5
Variance Ratios Test for Weekly Size-sorted Portfolio Returns with Equal Weights

	$q = 2$	4	8	26	52	156	260
Panel A. Portfolio of firms with market values in the smallest quintile (rank1)							
1977/8/31-2007/6/27	1.24	1.52	1.74	2.09	2.11	2.43	2.27
$T=1,541$	(0.000) **	(0.000) **	(0.000) **	(0.000) **	(0.000) **	(0.000) **	(0.008) **
	[0.000] **	[0.000] **	[0.000] **	[0.001] **	[0.003] **	[0.013] *	[0.037] *
1977/8/31-1992/8/26	1.27	1.62	1.95	2.55	3.16	4.43	4.05
$T=772$	(0.000) **	(0.000) **	(0.000) **	(0.000) **	(0.000) **	(0.000) **	(0.000) **
	[0.000] **	[0.000] **	[0.000] **	[0.001] **	[0.002] **	[0.009] **	[0.015] *
1992/8/26-2007/6/27	1.20	1.45	1.63	1.85	1.60	1.70	1.54
$T=769$	(0.004) **	(0.000) **	(0.000) **	(0.001) **	(0.048) *	(0.104)	(0.215)
	[0.000] **	[0.000] **	[0.000] **	[0.001] **	[0.042] *	[0.113]	[0.181]
Panel B. Portfolio of firms with market values in the middle quintile (rank3)							
1977/8/31-2007/6/27	1.11	1.26	1.34	1.39	1.28	1.63	1.84
$T=1,541$	(0.011) *	(0.001) **	(0.004) **	(0.032) *	(0.158)	(0.075)	(0.062)
	[0.000] **	[0.000] **	[0.000] **	[0.009] **	[0.101]	[0.072]	[0.078]
1977/8/31-1992/8/26	1.08	1.27	1.35	1.41	1.63	2.30	2.15
$T=772$	(0.127)	(0.020) *	(0.042) *	(0.108)	(0.077)	(0.027) *	(0.079)
	[0.018] *	[0.000] **	[0.002] **	[0.038] *	[0.037] *	[0.037] *	[0.082]
1992/8/26-2007/6/27	1.11	1.23	1.32	1.37	1.06	1.19	1.29
$T=769$	(0.038) *	(0.017) *	(0.022) *	(0.084)	(0.432)	(0.364)	(0.335)
	[0.001] **	[0.001] **	[0.003] **	[0.050] *	[0.374]	[0.289]	[0.252]

(continued)

TABLE 5 - Continued

	$q = 2$	4	8	26	52	156	260
Panel C. Portfolio of firms with market values in the largest quintile (rank5)							
1977/8/31-2007/6/27	1.00	1.06	1.10	1.18	1.20	1.63	1.88
$T=1,541$	(0.513)	(0.178)	(0.166)	(0.161)	(0.209)	(0.062)	(0.045) *
	[0.518]	[0.094]	[0.090]	[0.112]	[0.165]	[0.073]	[0.073]
1977/8/31-1992/8/26	0.97	1.07	1.11	1.25	1.52	2.44	2.52
$T=772$	(0.682)	(0.265)	(0.254)	(0.194)	(0.092)	(0.008) **	(0.020) *
	[0.778]	[0.148]	[0.143]	[0.119]	[0.061]	[0.029] *	[0.052]
1992/8/26-2007/6/27	1.00	1.03	1.08	1.11	0.90	0.84	0.84
$T=769$	(0.465)	(0.347)	(0.266)	(0.320)	(0.620)	(0.621)	(0.595)
	[0.456]	[0.310]	[0.223]	[0.284]	[0.579]	[0.525]	[0.463]

All values in the table are calculated from variance ratios for weekly portfolio returns with complete individual return histories from August 31, 1977 to June 27, 2007 ($T=1,541$). The values within parentheses are p-values from Lo and MacKinlay's (1988) heteroskedastic consistent test. The values within square brackets are p-values from Kan's (2006) exact test under the elliptical distribution assumption. * and ** indicate that the corresponding variance ratios are statistically different from 1 at the 5% and 1% significance level, respectively.

TABLE 6
Summary Statistics of Variance Ratios for Size-sorted Portfolios

number of portfolios	number of stocks	<i>q</i> -weeks						
		2	4	8	26	52	156	260
Panel A. median								
1	1,147	1.14	1.33	1.44	1.59	1.52	1.95	2.13
5	229	1.11	1.26	1.34	1.39	1.28	1.63	1.88
50	22	1.09	1.20	1.26	1.30	1.23	1.60	1.75
229	5	1.03	1.09	1.11	1.17	1.10	1.27	1.31
286	4	1.02	1.07	1.09	1.11	1.07	1.21	1.25
382	3	1.00	1.04	1.05	1.05	0.99	1.11	1.14
573	2	0.98	0.99	0.98	0.97	0.92	0.98	0.98
1,147	1	0.95	0.93	0.90	0.84	0.78	0.78	0.75
Panel B. VR < 1								
1	1,147	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
5	229	20.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
50	22	8.00%	4.00%	4.00%	2.00%	2.00%	0.00%	0.00%
229	5	24.45%	12.66%	11.35%	18.34%	36.24%	17.90%	17.90%
286	4	32.87%	17.13%	17.83%	24.83%	37.06%	24.13%	23.78%
382	3	47.12%	29.32%	31.15%	37.70%	51.05%	35.60%	35.86%
573	2	69.28%	56.72%	58.12%	58.99%	64.22%	52.88%	51.83%
1,147	1	86.14%	83.96%	82.82%	81.60%	82.74%	76.81%	73.58%

All values in the table are calculated from variance ratios for weekly portfolio returns with complete individual return histories from August 31, 1977 to June 27, 2007 ($T=1,541$). %(VR<1)'s represent percentages where variance ratios are less than one.

FIGURE 1
Sample Medians of Variance Ratios for Weekly Individual Stock Returns

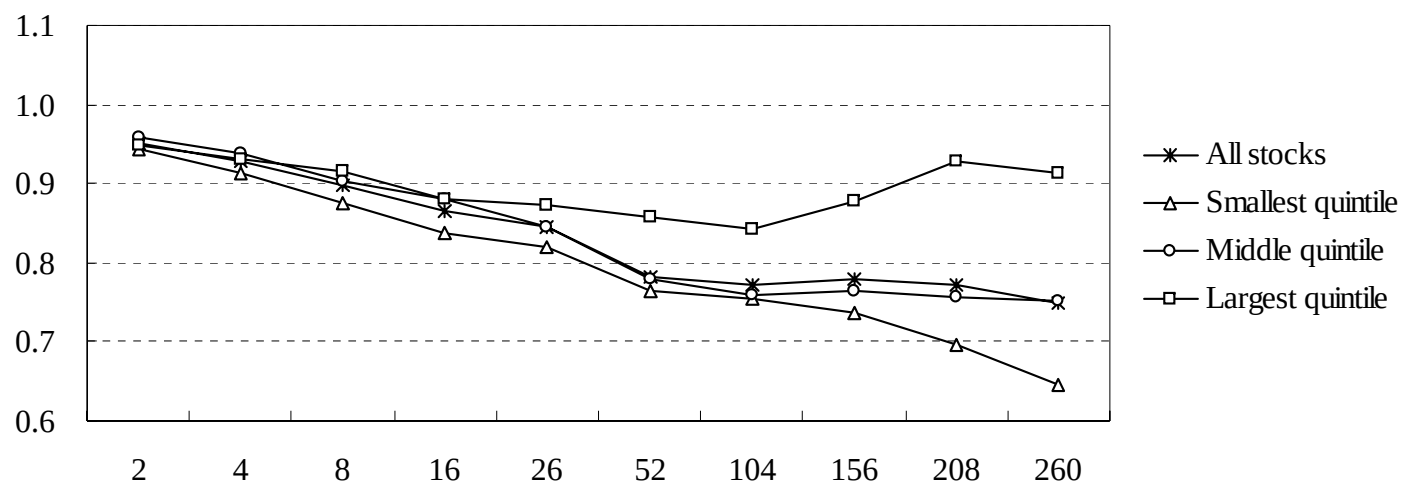


FIGURE 2
Sample Medians of Variance Ratios for Size-sorted 229 Portfolio Returns

